

A Unified Model of Game Design, Through the Lens of User Experience

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ABSTRACT

In this paper, two cornerstone game design and research models — the Mechanics, Dynamics, Aesthetics framework and the magic circle — are revisited and inspected through a lens of user experience design. This suggests two aspects that are missing from these models: a clear distinction between the intended and actual player experience, and the inclusion of transfer of knowledge, actions, and emotions between the game world and the real world, as part of the player experience. Based on these observations, a new unified model is proposed.

CCS CONCEPTS

• **Human-centered computing** → HCI theory, concepts and models; • **Applied computing** → Computer games; • **Software and its engineering** → Virtual worlds software.

KEYWORDS

game design, game analysis, game user experience

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1 INTRODUCTION

The rapidly evolving fields of game studies and game design have attracted the attention of numerous scholars, industry professionals, and students alike. As a result, several theories and models on the design and development of games have been proposed, inspired by various fields such as psychology, design, human-computer interaction, and computer science. While the technology and aesthetics of games themselves have changed a lot over the years, the scientific and practical models that we use to describe and study them remain largely relevant and accurate to this day. In this paper I will revisit two of these cornerstone models of game design and analysis: the Mechanics, Dynamics, Aesthetics (MDA) framework [17], and the magic circle [16]. Neither of these models distinguishes between the

intended player experience as it is crafted by the game designers, and the actual player experience as it occurs when a player starts interacting with the game in their particular context of use. In addition, the models do not attempt to divide the (intended and actual) player experience into its constituent parts, in order to create a deeper understanding of why games are experienced in a certain way by a particular player. After revisiting the MDA-framework and magic circle, and addressing key topics of debate, the two models will be combined by taking a user experience-focused approach, resulting in a unified representation of what happens when a player interacts with a game, and to what extent game designers can control what this experience is like.

Instead of adding new information or theories to the field, this paper aims to revisit and unify existing concepts and models, that until now were discussed in isolation. The resulting integrated model will be introduced in the following sections, and then presented in Fig. 4 towards the end of the paper. This proposed model can be used as a tool for dissecting games or design processes, in order to create an overview of how a game is designed to provide a particular player experience, and to identify whether this intended player experience matches the actual experience of a player, in their particular context of use. In addition, the model serves to communicate four important messages related to game design and the player experience: (1) there is a difference between the intended player experience (controlled by game designers), and the actual player experience (subject to context of use); (2) a game is a dynamic system, and the player experience results from interacting with this system as a whole, rather than individual game elements; (3) the boundary between the virtual game world and the real world can be seen as an interface, through which knowledge, behavior, and emotions can travel; (4) the player experience results from interacting with the game, subsequent reflection, and transfer (of knowledge, emotions) between the virtual and the real world. In its current state, the proposed model is based solely on theorycrafting and a synthesis from existing literature. It would therefore benefit from thorough discussion in the scientific and industry-based game design and research communities. This paper serves as a platform to start the conversation.

2 MECHANICS, DYNAMICS, AESTHETICS

The Mechanics, Dynamics, Aesthetics (MDA) framework is introduced as a tool for understanding and analyzing games, aimed at game designers and developers, critics, and researchers [17]. It breaks down the design and play of games into three components. From the player's perspective, an enjoyable experience emerges from interacting with a system, which consists of rules. From the designer's perspective, the rules are referred to as mechanics



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(components of the game), which create dynamics when they are combined into one system, and start interacting with each other and are interacted with by the player. These interactions result in the game's aesthetics, the (emotional) responses it triggers in the player. It is important to highlight the relationship between the parts, or design elements, that make up a game (referred to as mechanics in the MDA-framework), and how a dynamic system is created by combining these elements and having them interact with each other and with the player. It is this dynamic system that the player ends up interacting with and, as also noted by Hunicke et al. [17], it is challenging to predict the actual experience of interacting with the game because the dynamic system affords many different play styles, choices, and possibilities for interactions. As we move from individual game elements to an integrated, dynamic system, we may potentially also introduce issues that only emerge once the game's elements start interacting. An example of such an issue is *ludonarrative dissonance* [3, 14], a mismatch between the message that is conveyed through the narrative game elements, and the game play afforded by game elements relating to the mechanics of the game. For example, the story could portray the main character as a pacifist, yet the game play could invite the player to act aggressively. Although there is no issue with the elements when examined separately, introducing both to the same dynamic system could be detrimental to the player experience if the player values ludonarrative consistency.

Researchers have started discussing several potential limitations of the MDA-framework (e.g., [37]), mainly relating to the restrictive definition of 'mechanics' that seemingly ignores other important game elements, such as narrative aspects, and the fact that there is no direct link between the mechanics and aesthetics of the model. As a result, several extensions and modifications of the MDA-framework have been proposed [24, 27, pp. 41–43], that represent the story or narrative, as well as the technology that is used to implement the game as their own elements in the model. The criticism of the MDA-framework appears to stem from the fuzzy definitions of terminology such as 'mechanics' and 'aesthetics'. Mechanics have been defined solely as the 'verbs' of the game (cf. Sicart [29]), or all of the ways in which the player is able to interact with the game world: they press a button on the controller, and as a result the character jumps. However, from the way mechanics are further described in the MDA-framework it seems that the authors were referring to game elements in a broader sense, similarly to how Schell [27] groups several game element types under the term 'mechanics'. By broadening the definition of mechanics to include other game elements such as the virtual space or environment, objects, rules, goals, challenges, story, and user interface, the concept of 'dynamics' represents the entirety of a game system, consisting of individual game elements that interact with each other. Technology is then implicitly present in the way the designers have chosen to implement the game design elements, and combine them into the dynamic system. While it is possible for certain game elements to be more dominant and visible in the entirety of the dynamic system, the player experience can only be studied at the system level, not at the individual game element level, as the player interacts with the system as a whole.

A second point that is made by the authors of the MDA-framework is that there are two ways of looking at games: from

the perspective of game design studios as creators, and from the perspective of players as consumers of these games. However, it is unclear whether the 'aesthetics' in the model refer to the player experience as it is shaped by the designers, or the experience a player has as they play the game. This distinction between the creator and consumer of a game shares a resemblance with the concept of a *system image*, which was originally described by Norman in the context of product usability, affordances, and mental models [23, pp. 75–76], and later extended by Hodent in the context of game design to also include player experience [15, pp. 15–16] (Fig. 1). This theory describes that the designers of a product (in this case, a game) have a certain mental image or conceptual model in mind of how the product should operate and what the experience of using the product should be like. This vision is embedded in the system image, the game, to the best of the designers' abilities. The player, in turn, interacts with this game based on their own knowledge (i.e., of the world and of related products) and expectations, in a specific context of use, which results in the actual experience of interacting with the game. This context of use includes factors that are outside of the designers' control, such as the player's expectations and past experiences, or their current physical, emotional, or social context (e.g., playing with friends). In other words, while game designers can shape an enjoyable intended player experience, and avoid usability issues (such as quests that cannot be completed) to increase the likelihood of a positive actual player experience, this remains subject to individual differences (cf., for example, Hassenzahl and Tractinsky's definition of user experience [13]).

In short, the MDA-framework illustrates two important lessons regarding game design: (1) games are complex, dynamic systems that consist of individual elements which, when they start interacting with each other and the player, can result in a multitude of (unpredictable) outcomes; (2) there is an intended player experience, controlled by the game designers as they combine game elements into a dynamic system, and an actual player experience that is partly outside of the designers' control. Fig. 2 presents a modified version of the MDA-framework which proposes a general definition of game elements instead of mechanics, illustrates how the dynamic system consists of game elements that interact with each other, and incorporates the difference between the intended and the actual player experience.

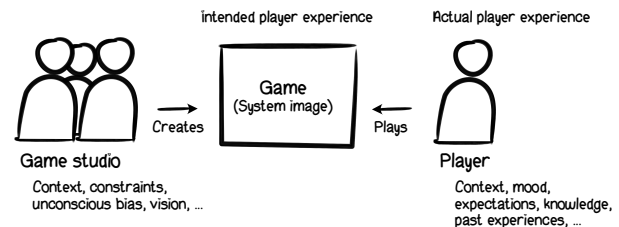


Figure 1: The game studio designs a system image (game) according to their frame of reference, providing an intended player experience, and the player has the actual player experience while playing the game, based on their frame of reference and context of use. Image based on the work of Norman [23] and Hodent [15].

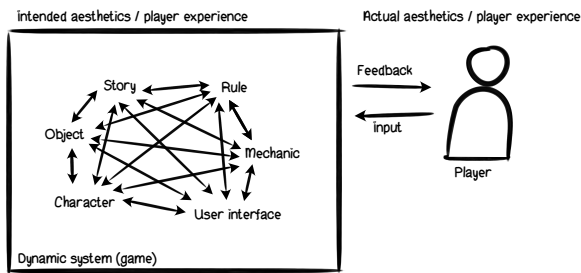


Figure 2: Reinterpretation of the MDA-framework [17]: A dynamic system (the game) emerges from interactions between game elements (‘mechanics’), the player experience results from a player’s interactions with this dynamic system and the player’s context.

3 MAGIC CIRCLE

The concept of a magic circle, an environment that is separate from our everyday lives in which players engage in game play, was first briefly introduced by Johan Huizinga as early as 1938 [16]. It has since been used as a metaphor to describe a game world that can contain different (and often artificial) rules and conflicts from what we know from the real world [26]: The act of shooting a ball into a goal holds no value in the real world, yet it grants the player a point in the game world of soccer. The game world is seen as a place where players tend to be safe from physical and emotional harm [26], making games an important medium for exploring and developing one’s abilities and identity, without serious real world risks or consequences. The magic circle can be demarcated by a physical arena (e.g., soccer field or chess board), but it can also exist digitally or in the minds of the players.

The magic circle has been a frequent topic of debate [5, 7, 18, 31], where the majority of the criticism relates to what Stenros calls the *strong boundary hypothesis*: the assumption that the boundary that separates the magic circle from the real world is clear, uniform, and ever-present [31]. This would mean that games can provide utopian environments that completely shut their players off from the real world and their everyday lives, ignoring the fact that players are still situated in the real world as they engage in game play [7]. As a result, the magic circle is considered a formalist tool that helps to decipher the structure of games, but fails to consider the context in which games are played. If the metaphor of the magic circle indeed does not allow for a clear separation between the real and the virtual, and does not consider the real world context of play, is it still a useful tool for studying games and player experience?

Especially when considering video games, it is apparent that there is no clear-cut boundary between the real world and the game world: Players can bring information into the game (e.g., walkthroughs) and acquire real world knowledge from the game, their in-game behavior can be based on real life characteristics or ambitions, with esports and live streaming it is possible make real world money, and real world social dynamics can carry over into multiplayer game sessions with friends. These elements (i.e., emotions, actions, knowledge) that *transfer* between the real world and the game world contribute toward a player’s actual player experience in the real world. By considering the boundary of the

magic circle as an *interface* rather than a clear separation, we can create an inventory of everything that passes through this interface to see how and why a particular player experience results from playing the game. This is similar to what Garfield describes as the metagame, or ‘how the game interfaces with life’ [10, p. 1]. This refers to knowledge and emotions the player brings into, and takes away from the game. The concept of the metagame reiterates the previously discussed point that the player experience is not only determined by the way the game is designed (e.g., the rules and interactions that take place), but also by the real world context from which an individual player approaches the game. Because this real world context is involved, what happens in the metagame – the interface between the game world and real life – is not fully under the designers’ control, although Salen and Zimmerman [26] note that it is possible, to some extent, to design for the metagame. They illustrate this with an example of Magic: The Gathering (Wizards of the Coast), initially put forth by Garfield, in which core activities such as building card decks and strategizing are purposefully designed to cross the boundary between play sessions in the magic circle and the real world.

Particularly in the context of video games, the metaphor of the magic circle might be more clear if we assume that players do not enter the magic circle when they engage in game play, but rather they interact with this virtual environment through a representation of the self within it: the avatar (Fig. 3). By introducing this separation between the player and the character or object they are controlling within the game world, we can distinguish between experiences that result from observing and interacting with this world from the real world, and information and emotions that are part of the metagame, that travel through the interface between the virtual and real worlds. Note that the visualization of the magic circle in Fig. 3 resembles the reinterpretation of the MDA-framework shown in Fig. 2, if the magic circle is considered to be defined and maintained by the dynamic system, the game.

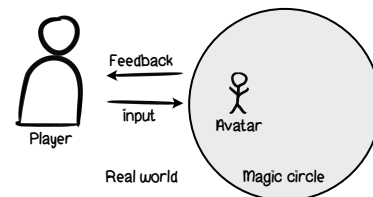


Figure 3: The magic circle [16, 26], of which the boundary is considered an interface between the real world and the magic circle (virtual game world). The player, situated in the real world, is represented by their avatar as their virtual self in the magic circle.

4 THE AESTHETIC DISTANCE AND TRANSFER

The degree to which there is a separation between the real world and the virtual world, or magic circle, can be described by the *aesthetic distance* [38, 40]. This distance is not dichotomous, but ranges from what Wolf calls ‘disinterested observation of an artefact’ at

the most distant level, and ‘complete immersion in the represented world’ at the least distant level. At a smaller aesthetic distance, the perceived difference between real world emotions and ideas, and events from the virtual world is blurred, which results in experiencing an *aesthetic illusion* [40]. The aesthetic illusion is related to concepts such as suspension of disbelief, transportation and immersion or presence. Although immersion and presence are often used interchangeably, Slater [30] suggests a distinction between design elements that stimulate feelings of being there in the virtual world (immersion), and the actual feeling of being there as perceived by the player (presence). This is similar to the difference between the intended player experience and the actual player experience presented in the previous sections of this paper. For example, designers could make a racing simulator take place in virtual reality, using a steering wheel input device, which can be seen (objectively) as a highly immersive game. At the same time, if the player is constantly distracted by external factors such as a noisy environment or stress unrelated to the game, the resulting (subjective) feeling of presence could still be low. Also from a narrative perspective, designers could foster immersion by creating relatable characters and a compelling story, after which presence is determined by an individual player’s ability to identify with the protagonist and become fully involved with the story, which could differ between players even though the designed level of immersion is identical. Immersion is related to realism, in terms of graphics, story elements, and the behavior of characters and objects [2, 12, 25, 28], although it does not appear to be fully clear if realism boosts immersion, immersion increases perceived realism, or both.

The effect of a high degree of presence, or a small aesthetic distance, is two-fold. Mekler et al. [21] conducted a systematic review on game enjoyment, which they consider an important component of player experience. They conclude that enjoyment can be seen as the valence of the player experience (either pleasant or unpleasant), while immersion (presence) determines the intensity of this experience. In other words, if a player feels more present, the boundary between the real and virtual worlds will become thinner (aesthetic distance is smaller), and the player experience that results from interacting with the game becomes more intense. Games that — purposefully or inadvertently — have a larger aesthetic distance can provide relief, while increased presence can lead to achieving a flow state [21]. The second effect of a small aesthetic distance is related to the aforementioned transfer of knowledge, skills, and emotions from the virtual game world into the real world. Serious games such as training simulations [2] or persuasive games and narratives [11, 35] rely on transfer in order to achieve real world outcomes, which means that a small aesthetic distance is a requirement for them to be successful. Transfer also occurs in entertainment games, although it may not be the primary intention of their designers. For example, Car Mechanic Simulator (Red Dot Games), due to the realistic way in which the inner workings of a car are represented, can teach its player about the different components of a car and how they relate to each other. Transfer can also occur from the real world into the game world, for example if a player chooses to respond to a game’s dialogue options based on what their real life response would be, or if they apply a strategy that was found online. Both forms of transfer (in and out of the game) can help to create a more engaging and meaningful player experience.

A player can experience different types of presence [19]: physical presence, which describes the feeling of being there in the virtual environment, and interacting with virtual objects; social presence, related to the interactions with (artificial and real) others; and self-presence, the ability of being yourself in the virtual environment, with no distance between the player and their avatar. Busselle and Bilandzic add the concept of narrative presence [6], which describes the feeling of being part of the story, and empathizing with the characters involved. These different types of presence can lead to different types of transfer. Physical, social, and narrative presence relate to observing interactions between various game elements: If the story is realistic and relatable, the player could learn how to respond to similar real world situations. Self-presence is different in that it does not reduce or remove the boundary between events that happen in-game and real world events, but rather between the identity of the player and their avatar. As a result, if the player feels highly self-present, they might experience actions and dialogue from the game as being directed at them instead of their avatar. In other words, instead of strongly empathizing with their avatar in the virtual world as is the case with narrative engagement, they are directly feeling what their character is feeling (e.g., guilt, love). This is known in live action role playing (LARP) as *bleed* [4, 22], where players feel a great degree of self-presence with the character they are portraying, and as a result they truly experience events such as the loss of another character as if they occurred in real life, resulting in real and intense emotional responses. This effect has also been studied in digital role playing games with non-playable characters, where players indicated genuine feelings of love for an in-game character [36]. Bleed is also a common phenomenon in massively multiplayer online role playing games (MMORPGs), in which players can communicate out of character and thereby form real life relationships [39]. Similar to the other forms of transfer, bleed can happen in both directions. There can be bleed-in, for example when a player acts differently towards another player because of their relationship outside of the game, and bleed-out when a player experiences emotions that were targeted at their avatar instead. Bleed can be a powerful, albeit risky tool to enable a player to experience events or feelings they might not (safely) be able to experience in real life. Also in this case, the degree of experienced bleed is a result of combined efforts by the game designers (e.g., creating a more relatable character to stimulate bleed) and the players, who can choose to either welcome or discourage bleed from happening, for example by choosing whether or not to draw character traits from their real world situation [4].

Two final types of transfer happen on a subconscious level and have not received much attention yet in game studies: the *aesthetic experience* [8, 20] and *transformative experience* [9, 32, 33]. An aesthetic experience occurs when a game is not simply observed, but actively experienced, which can result in an “exceptional state of mind” [20, p. 1] and is “[...] an immersion so complete that the qualities of the object and the emotions it arouses have no separate existence” [8, p. 288]. It is a way of communicating with the designers, and their culture, through the game as an expressive object [8], and placing the meaning conveyed by the object within the player’s past experiences and world views. Transformative experiences include feelings such as awe, and result in a change in the player’s world views. Having an aesthetic or transformative

experience is (slightly) different from reflecting on a play experience, as it is a direct subconscious emotional response, rather than conscious processing of an interaction, and experiencing (emotional) responses as a result of this reflection. This form of transfer, like the other components of the player experience, is highly individualistic [9, 34], as players will have different responses to the message that is conveyed through the game, and how this aligns with their existing world views. Having such an experience can lead to a new understanding of the self and of one's reality.

The feeling of presence, and subsequent transfer, can change through time, even within a single play session, and experiencing one type of presence (e.g., self-presence) does not automatically mean that other types are experienced as well. For example, players can talk out of character in an MMORPG (high self-presence), while completing quests that have no effect on the real world (low physical and narrative presence) and merely act as a relaxing task while chatting. It is also possible to experience different ways of transfer, even within one type of presence. The feeling of being physically present in Car Mechanic Simulator supports transfer of knowledge about car components, but the simulation of actually working on the car (the mechanics of the game) is less realistic (e.g., mouse clicks to remove and mount parts), therefore the skills to conduct repairs on a real car do not transfer.

5 TOWARDS A UNIFIED MODEL

The previous sections examined the MDA-framework and the magic circle through the lens of user experience, by adding a distinction between the intended player experience in the virtual world, which is provided by the dynamic game system built by the game designers, and the actual player experience as it occurs in the real world. This player experience, particularly the intensity of it and how much transfer occurs, is determined by the — deliberately or unintentionally introduced — immersive design elements, and the player's openness and ability to feel present in the virtual environment. This degree of presence, or the aesthetic distance, determines the boundary between the virtual and the real worlds. The model displayed in Fig. 4 combines these insights from revisiting the MDA framework and magic circle. It can be used as a tool to analyze the intended player experience, implemented as the dynamic system. In addition, designers and researchers can use the model to create an inventory of key determinants of the actual player experience. These determinants include the player's contextual factors, interactions with the game and how they might trigger reflective thinking, the aesthetic distance and how it is determined by immersive design and the player's likelihood to feel presence, bleed between the player and their avatar, and any transfer of information, emotions, and actions that might occur between the real world and the virtual world. Finally, the model serves as a reminder that (1) the actual player experience is only partly determined by a game's design; (2) a game is more than the sum of its parts; (3) transfer between the real and virtual worlds occurs based on presence; and (4) presence and transfer influence the actual player experience.

In future work, it would be interesting to look at the related domains of free-form play, and gameful or playful interactions such as gamification. Do these also rely on bleed and transfer, or is generating engagement through providing enjoyable interactions

enough? In addition, while the model could help to create a clear overview of the player experience and its components (e.g., how much transfer occurs), it does not offer a way of actually measuring the quality of the player experience, and how one particular experience compares to another. It would be interesting to see to what extent player experience evaluation tools such as the Player Experience Inventory [1] align with the model presented in this paper. In supplementary materials, I have made a first attempt at describing my player experience using the different elements of the unified model¹. Finally, the model does not yet take into account differences in player types or personalities. Perhaps there is a difference in players' needs regarding the intensity of their experience, where some players actively seek out games that offer transfer or reflection, while others prefer to avoid them.

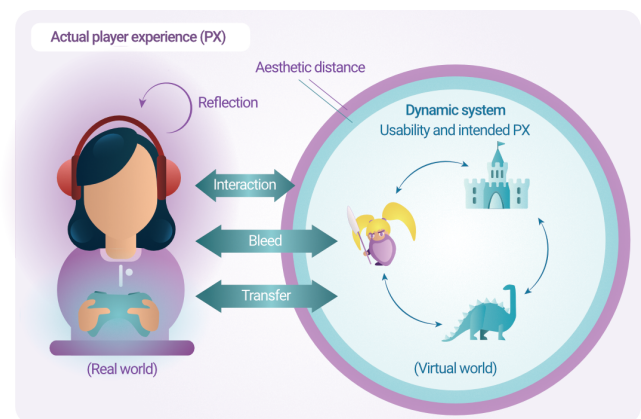


Figure 4: The unified model combines the MDA-framework [17] and the magic circle [16, 26]. The player, situated in the real world, interacts with a dynamic system that defines a virtual world (magic circle), with an intended player experience. The player can observe and interact with the virtual world, and reflect based on these interactions. If the aesthetic distance is small, knowledge, emotion, and behavior can transfer between the real and virtual worlds, leading to a more intense and meaningful player experience.

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